Physiology of Blood: Hemoglobin, Anemia, & Polycythemia

1. Hemoglobin synthesis begins early in the erythroid lineage. During the chemical assembly of the heme moiety, the very first step occurs when two molecules of succinyl-CoA combine with two molecules of glycine to form:
a. Protoporphyrin IX
b. A pyrrole ring
c. Free iron complexes
d. Alpha-amino-beta-ketoadipic acid (which converts to delta-aminolevulinic acid)
e. A single globin chain
2. Four individual pyrrole rings subsequently coalesce to construct a complex heterocyclic ring structure known to physiology as:
a. Biliverdin
b. Urobilinogen
c. Protoporphyrin IX
d. Apoferritin
e. Transferrin
3. Once protoporphyrin IX is synthesized, it combines chemically with iron to form a complete heme molecule. Which specific state of iron is strictly required for functional binding to protoporphyrin IX?
a. Ferrous state (Fe^{2+})
b. Ferric state (Fe^{3+})
c. Metallic iron (Fe^0)
d. Ferritin complexes
e. Hemosiderin matrices
4. Each completed molecule of normal adult Hemoglobin A (HbA) contains a total protein component composed of four distinct polypeptide globin chains, designated as:
a. Two alpha and two gamma chains
b. Two alpha and two beta chains
c. Two beta and two delta chains
d. Four alpha chains purely
e. Four beta chains exclusively
5. Because a single molecule of adult Hemoglobin A contains four independent heme groups, each fully saturated hemoglobin molecule can transport a maximum of how many individual atoms of oxygen?
a. Two atoms
b. Four atoms
c. Eight atoms (forming four O_2 molecules)
d. Twelve atoms
e. One single atom
6. The most crucial physiological feature of hemoglobin-oxygen binding is that it is loose and completely reversible. The specific chemical component of the hemoglobin molecule that binds directly with a molecule of oxygen is the:
a. Ferrous iron atom (Fe^{2+}) within the heme ring
b. Imidazole ring of histidine
c. Amino terminal of the alpha globin chain

d. Carboxyl terminal of the beta globin chain
e. Protoporphyrin side chain

7. When oxygen binds reversibly to the ferrous iron in hemoglobin, it does not oxidize the iron atom. Instead, this gas transport phase is classified as a process of:

a. Oxidation
b. Reduction
c. Oxygenation
d. Carbonylation
e. Hydrolysis

8. In fetal life, the primary hemoglobin manufactured by the fetus exhibits a higher affinity for oxygen than adult hemoglobin, facilitating oxygen transfer across the placenta. This is Fetal Hemoglobin (HbF), which consists of:

a. Two alpha and two beta chains
b. Two beta and two delta chains
c. Four gamma chains exclusively
d. Two alpha and two gamma chains
e. Two alpha and two delta chains

9. A single point mutation causing the substitution of valine for glutamic acid at the sixth position of the beta globin chain produces an abnormal variant of hemoglobin known as HbS. Under low oxygen tension, this hemoglobin precipitates, causing:

a. Thalassemia major
b. Hereditary spherocytosis
c. Pernicious anemia
d. Aplastic anemia
e. Sickle cell anemia

10. When old or fragile erythrocytes are systematically destroyed by the macrophages of the reticuloendothelial system, the globin portion of hemoglobin is broken down into amino acids, and the heme ring is cleaved to release free iron and a green pigment called:

a. Bilirubin
b. Urobilinogen
c. Hemosiderin
d. Stercobilin
e. Biliverdin

11. The green biliverdin is rapidly reduced by macrophages into a yellow-orange lipid-soluble pigment that is released into the blood plasma as:

a. Free (unconjugated) bilirubin
b. Conjugated bilirubin diglucuronide
c. Urobilin
d. Cholecystokinin
e. Haptoglobin

12. Because free bilirubin is highly insoluble in water, it cannot travel naked through the blood. To reach the liver safely, it binds tightly to which circulating plasma protein?

a. Fibrinogen
b. Transferrin
c. Alpha globulin
d. Prothrombin

e. Plasma albumin

13. Within the liver, hepatocytes detach free bilirubin from its transport protein and conjugate it primarily with glucuronic acid to form water-soluble bilirubin glucuronide, which is then excreted into the:

a. Intestinal tract via the bile
b. Systemic arterial circulation
c. Urinary bladder
d. Pericardial cavity
e. Pancreatic duct lumen

14. Anemia is fundamentally defined as a pathologically diminished concentration of hemoglobin in the blood. When classifying anemia morphologically, the calculation used to determine the average volume of an individual red blood cell is the:

a. Mean Corpuscular Hemoglobin (MCH)
b. Mean Corpuscular Hemoglobin Concentration (MCHC)
c. Packed Cell Volume (PCV)
d. Mean Corpuscular Volume (MCV)
e. Total Erythrocyte Mass

15. To differentiate whether a red blood cell has a normal, concentrated, or diluted amount of hemoglobin per unit volume, clinicians calculate the:

a. Mean Corpuscular Volume (MCV)
b. Mean Corpuscular Hemoglobin Concentration (MCHC)
c. Total Blood Volume
d. Color Index purely
e. Reticulocyte Production Index

16. An anemia characterized by a normal cell size (normal MCV) and a normal hemoglobin concentration (normal MCHC), typically seen immediately following an acute hemorrhage, is classified as:

a. Microcytic, hypochromic anemia
b. Normocytic, normochromic anemia
c. Macrocytic, hypochromic anemia
d. Megaloblastic anemia
e. Microcytic, normochromic anemia

17. Chronic dietary iron deficiency or a failure of iron absorption severely impairs hemoglobin synthesis. This produces a classic morphological pattern of anemia characterized as:

a. Macrocytic, hyperchromic
b. Normocytic, normochromic
c. Microcytic, hypochromic anemia
d. Megaloblastic
e. Macrocytic, normochromic

18. A failure of normal DNA replication and nuclear maturation due to a deficiency of either Vitamin B12 or Folic acid causes erythroblasts to grow large and fragile, resulting in a morphological classification of:

a. Macrocytic (megaloblastic) anemia
b. Microcytic anemia
c. Normocytic anemia
d. Sickle cell anemia
e. Microcytic hypochromic anemia

19. When classifying anemia etiologically (based on the underlying cause), a complete failure or structural destruction of the bone marrow stroma due to radiation, toxic chemicals, or autoimmune injury leads to:
a. Hemolytic anemia
b. Aplastic anemia
c. Pernicious anemia
d. Iron deficiency anemia
e. Erythroblastosis fetalis
20. Hereditary spherocytosis is an etiological type of hemolytic anemia. It is caused by a genetic defect in the red blood cell membrane structural proteins (such as spectrin), making the cells round, fragile, and vulnerable to destruction inside the:
a. Kidneys
b. Spleen (via splenic macrophage gating)
c. Bone marrow capillaries
d. Pulmonary circulation
e. Hepatic veins
21. In Thalassemia, a genetic defect impairs the synthesis rate of either the alpha or beta globin polypeptide chains. Etiologically, this results in a combination of defective hemoglobin production and severe:
a. Autoimmune aplasia
b. Megaloblastic macrocytosis
c. Acute post-hemorrhagic shock
d. Gastrointestinal blood loss
e. Extravascular and intravascular hemolysis (hemolytic anemia)
22. A patient with a severe, chronic microcytic hypochromic iron-deficiency anemia will display a highly unique diagnostic laboratory profile, featuring:
a. High serum ferritin and low total iron-binding capacity (TIBC)
b. Low serum iron, low serum ferritin, and markedly elevated total iron-binding capacity (TIBC)
c. Elevated MCV and elevated MCHC
d. Normal hematocrit with an elevated reticulocyte count
e. High transferrin saturation with low erythropoietin
22. Under normal physiological conditions, what is the standard range for the Mean Corpuscular Volume (MCV) of human erythrocytes?
a. 50 to 60 fL
b. 120 to 140 fL
c. 80 to 100 fL (femtoliters)
d. 10 to 20 fL
e. 150 to 200 fL
24. The primary physiological consequence of any severe anemia is a profound reduction in the oxygen-carrying capacity of the blood. To compensate for this tissue hypoxia, the cardiovascular system rapidly executes:
a. Severe peripheral vasoconstriction to lower heart rate
b. A total shutdown of renal blood flow
c. An extreme elevation of systemic vascular resistance
d. A marked increase in cardiac output and a decrease in blood viscosity
e. Immediate bradycardia

25. Polycythemia is defined as an abnormal increase in the total mass of circulating red blood cells in the vascular system. Polycythemia vera (erythremia) is a primary polycythemia caused by a:
a. Chronic hypoxic environment
b. Clonal, autonomous tumor-like aberration in the hematopoietic organs
c. High-altitude atmospheric shift
d. Severe state of systemic dehydration
e. Massive hypersecretion of renal erythropoietin
26. In a patient with polycythemia vera, the red blood cell count can rise to 7 to 8 million per cubic millimeter, and the hematocrit can reach 60% to 70%. This massive cell overload causes a dramatic increase in blood:
a. Velocity through the microvasculature
b. Alkalinity and pH
c. Viscosity (sometimes up to three to five times the viscosity of normal water)
d. Temperature
e. Electrical conductivity
27. Secondary polycythemia occurs as a normal physiological adaptation driven by a hormone pathway. It is triggered when tissues experience chronic hypoxia, stimulating the hypersecretion of:
a. Thrombopoietin by the liver
b. Interleukin-1 by macrophages
c. Cortisol by the adrenal cortex
d. Erythropoietin (EPO) by the kidneys
e. Growth hormone by the pituitary
28. A common physiological form of secondary polycythemia occurs in normal populations who live at highly elevated altitudes (above 11,000 to 14,000 feet), a condition termed:
a. Polycythemia vera
b. Relative polycythemia
c. Biospheric polycythemia
d. High-altitude polycythemia
e. Gaisböck’s syndrome
29. Secondary polycythemia can also occur in individuals with chronic cardiac or respiratory disease (such as severe lung failure or congenital right-to-left cardiac shunts) because these pathologies directly cause:
a. Total marrow aplasia
b. Accelerated destruction of transferrin
c. Tissue hypoxia due to poor oxygenation of systemic arterial blood
d. Excessive production of intrinsic factor
e. Hyper-absorption of dietary iron salts
30. Relative polycythemia (spurious polycythemia) differs completely from primary and secondary types because the total body red blood cell mass remains normal. Instead, the hematocrit appears elevated secondary to a profound loss of:
a. Plasma volume (such as from severe dehydration, sweating, or severe burns)
b. Bone marrow stem cells
c. Hemoglobin beta chains
d. Spleen macrophage capacity
e. Total blood leukocytes

31. In polycythemia vera, because the capillary beds become heavily engorged with highly viscous, slow-moving blood, a large fraction of the hemoglobin loses its oxygen to the tissues, manifesting clinically as a purple skin discoloration called:
a. Jaundice
b. Pallor
c. Erythema
d. Cyanosis
e. Albinism
32. Carbon monoxide (CO) gas poses a fatal hazard because it binds to the iron atom of hemoglobin at the exact same site as oxygen, but with an affinity that is approximately:
a. 2 times weaker than oxygen
b. 10 times stronger than oxygen
c. 250 times stronger than oxygen
d. Equal to oxygen affinity precisely
e. 1,000,000 times weaker than oxygen
33. When carbon monoxide binds to one of the four heme groups in a hemoglobin molecule, it shifts the oxygen-hemoglobin dissociation curve to the left, which has the dangerous physiological effect of:
a. Preventing the other three hames from binding oxygen at all
b. Markedly inhibiting the release of the remaining bound oxygen to the tissues
c. Accelerating the hepatic clearance of iron
d. Triggering immediate microcytic aplasia
e. Instantly destroying the erythrocyte membrane
34. The unique molecular configuration that allows hemoglobin to transport oxygen without permanently binding it is due to the iron atom being coordinated within the porphyrin ring and legally guarded by an adjacent amino acid residue named:
a. Tryptophan
b. Alanine
c. Valine
d. Proximal and distal Histidine
e. Leucine
35. If the ferrous iron (Fe^{2+}) in hemoglobin is pathologically oxidized to the ferric state (Fe^{3+}), the molecule loses its capacity to bind oxygen entirely. This oxidized, non-functional variant of hemoglobin is:
a. Carboxyhemoglobin
b. Carbaminohemoglobin
c. Methemoglobin
d. Sulfhemoglobin
e. Oxyhemoglobin
36. A defects-based etiological type of hemolytic anemia caused by an inherited deficiency of the enzyme glucose-6-phosphate dehydrogenase (G6PD) makes erythrocytes highly sensitive to oxidative stress from certain drugs or foods, leading to the precipitation of hemoglobin as:
a. Howell-Jolly bodies
b. Heinz bodies
c. Pappenheimer bodies
d. Cabot rings

e. Multilamellar bodies

37. When analyzing anemic states, a classic etiological cause of microcytic hypochromic anemia in developing regions is chronic, low-grade blood loss from the gastrointestinal tract due to a parasitic infection by:

a. Plasmodium falciparum
b. Hookworms (such as Ancylostoma duodenale)
c. Entamoeba histolytica
d. Taenia saginata
e. Ascaris lumbricoides

38. The normal concentration of hemoglobin in whole blood for a healthy adult male averages approximately:

a. 5 g/dL
b. 10 g/dL
c. 25 g/dL
d. 34 g/dL
e. 15 g/dL (typically 14 to 16 g/dL)

39. In a normal, healthy adult female, the standard expected concentration of hemoglobin per 100 mL of whole blood is roughly:

a. 8 g/dL
b. 14 g/dL (typically 13 to 15 g/dL)
c. 11 g/dL
d. 18 g/dL
e. 22 g/dL

40. Each individual gram of pure, functional hemoglobin is physiologically capable of binding chemically with a maximum volume of oxygen equal to:

a. 1.34 mL of oxygen
b. 0.5 mL
c. 5.0 mL
d. 15.0 mL
e. 22.4 mL

41. Anemia of chronic disease (seen in long-standing infections or malignancies) is an etiological type of anemia driven by inflammatory cytokines that stimulate the liver to overproduce a peptide hormone that blocks iron absorption and trapping, known as:

a. Transferrin
b. Erythropoietin
c. Hepcidin
d. Apoferritin
e. Biliverdin

42. Which diagnostic parameter measures the percentage of whole blood volume occupied by packed red blood cells, serving as a cornerstone for identifying anemia or polycythemia?

a. Mean Corpuscular Volume (MCV)
b. Mean Corpuscular Hemoglobin (MCH)
c. Reticulocyte Count
d. Hematocrit (Packed Cell Volume)
e. Erythrocyte Sedimentation Rate (ESR)

43. If a patient presents with a severe macrocytic anemia (MCV > 115 fL) accompanied by a normal MCHC, the clinician must immediately suspect a deficiency affecting:
a. Vitamin B12 or Folic Acid (maturation factor deficiency)
b. Iron availability or hepcidin levels
c. Structural cell wall spectrin
d. Alpha globin chain synthesis rates
e. Erythropoietin synthesis in the peritubular cells
44. Polycythemia vera is classified pathologically as a chronic myeloproliferative disorder. A highly high-yield genetic feature present in over 95% of patients with polycythemia vera is a somatic point mutation in the gene for:
a. Janus Kinase 2 (JAK2 V617F mutation)
b. Erythropoietin receptors
c. Hemoglobin beta chains
d. Spectrin anchoring fibers
e. Carbonic anhydrase enzymes
45. In high-altitude polycythemia, the blood volume expands alongside the red cell mass. Despite the increased blood viscosity, systemic tissue blood flow in these individuals is maintained close to normal because:
a. Systemic arterial blood pressure drops to near zero
b. The heart rate slows down to a resting state of 20 beats per minute
c. Hypoxia causes local autoregulatory vasodilation of peripheral blood vessels
d. Carbon dioxide levels in the blood completely disappear
e. The red blood cells lose their membranes to become smaller
46. When hemoglobin breaks down inside splenic macrophages, the iron atom is released into the intracellular pool. To be transported safely out of the macrophage back into the plasma via the ferroportin channel, the iron must combine with:
a. Apoferritin
b. Biliverdin reductase
c. Apotransferrin to enter the circulating transferrin pool
d. Albumin fragments
e. Heme oxygenase-1 complexes
47. An etiological type of anemia caused by an autoimmune reaction where antibodies are manufactured against the patient's own red blood cell surface antigens, leading to rapid cell clearance, is:
a. Autoimmune hemolytic anemia
b. Aplastic anemia
c. Pernicious anemia
d. Iron deficiency anemia
e. Thalassemia major
48. In sickle cell anemia, the distorted, rigid sickle cells become jammed inside the narrow capillaries when they lose oxygen. This creates a dangerous pathological cycle characterized by:
a. Rapid overproduction of white blood cells
b. Local tissue infarction, intense pain crises, and accelerated splenic hemolysis
c. Sudden conversion of hemoglobin into methemoglobin
d. Total suppression of kidney erythropoietin release
e. Massive hypertrophy of the left ventricle purely

49. Spurious or relative polycythemia can be triggered by a sudden, massive shift of fluid out of the intravascular space. This presentation is commonly seen in emergency clinical practice following:

a. Chronic ingestion of water
b. Profuse diarrhea, severe vomiting, or excessive administration of diuretic drugs
c. Living at sea level for twenty years
d. A highly successful blood transfusion matching the recipient
e. Total block of the hepcidin gene transcription

50. The precise structural mechanism that prevents hemoglobin from leaking out of the vascular system into the tissues under normal conditions is that it is entirely enclosed within the:

a. Endothelial cell junctions
b. Intracellular cytoplasm of circulating erythrocytes
c. Plasma albumin complexes
d. Connective tissue adventitia of large arteries
e. Fibrous trabeculae of the spleen

Answer Key Table (Hemoglobin, Anemia, & Polycythemia)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-50
d	e	c	b	b	c
c	e	a	c	d	d
a	a	b	d	c	a
b	a	d	c	b	a
b	b	b	c	b	c
a	b	c	c	e	c
c	a	d	c	b	a
d	b	d	a	a	b